

Localized PM2.5 Prediction Model for Mae Fah Luang University

Senior Project 1 Final Presentation by The Outliers

- Phyo Thant Kyaw
- Phone Wai Yan Thaing
- Kyaw Phone Thu
- Kyaw Thura

- **Advisor: Aj. Khwunta Kirimasthong**
- **Committee: Aj. Thongchai Yooyativong**
- **Committee: Aj. Paweena Suebsombut**

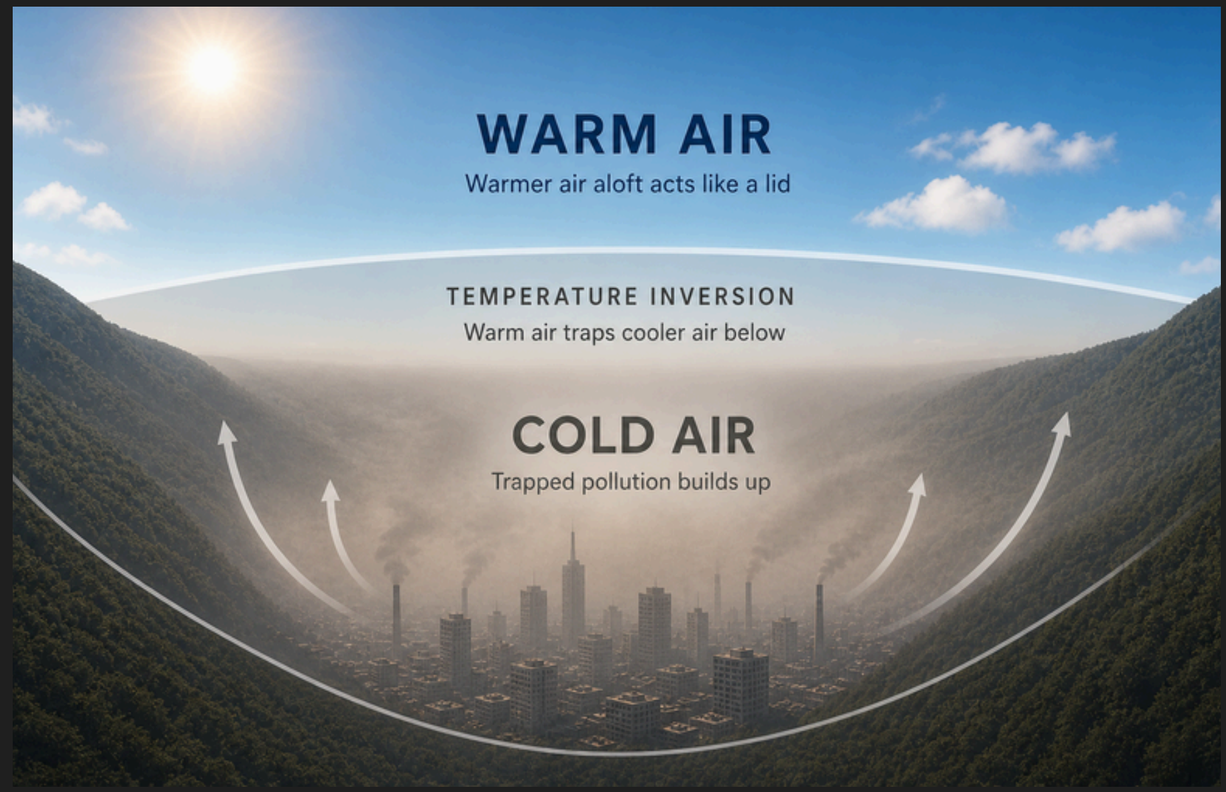
PROBLEM STATEMENT

The Haze Crisis: A Threat to Health in Northern Thailand

According To **Pollution Control Department (PCD) Thailand**

- The **Crisis**: Severe annual haze in Northern Thailand.
- The **Peak Season**: February to April.
- The **Impact**: Critical health risks for the MFU community (students & staff)

- **The Geography:** Deep valley basins ("The Valley Effect").
- **The Mechanism:** Temperature inversions act as a lid.
- **The Result:** A "**Bowl Effect**" that traps and concentrates hazardous PM2.5



Lack of localized early warnings leaves the MFU community vulnerable to respiratory health issues.

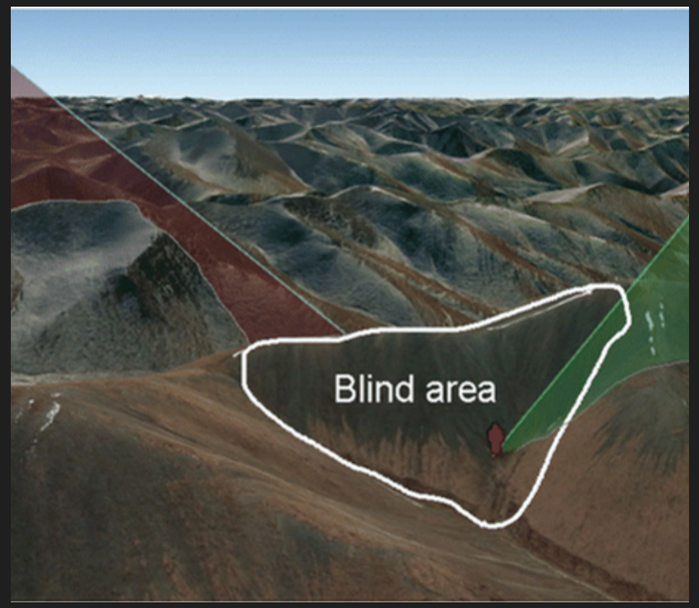
EXISTING SOLUTIONS

Identifying the Gap in Current PM2.5 Monitoring



Global Models & AirVisual

- Relies on generalized global data.
- **Coarse 9km grid resolution.**
- Too broad for campus-level accuracy



Satellite Limitations

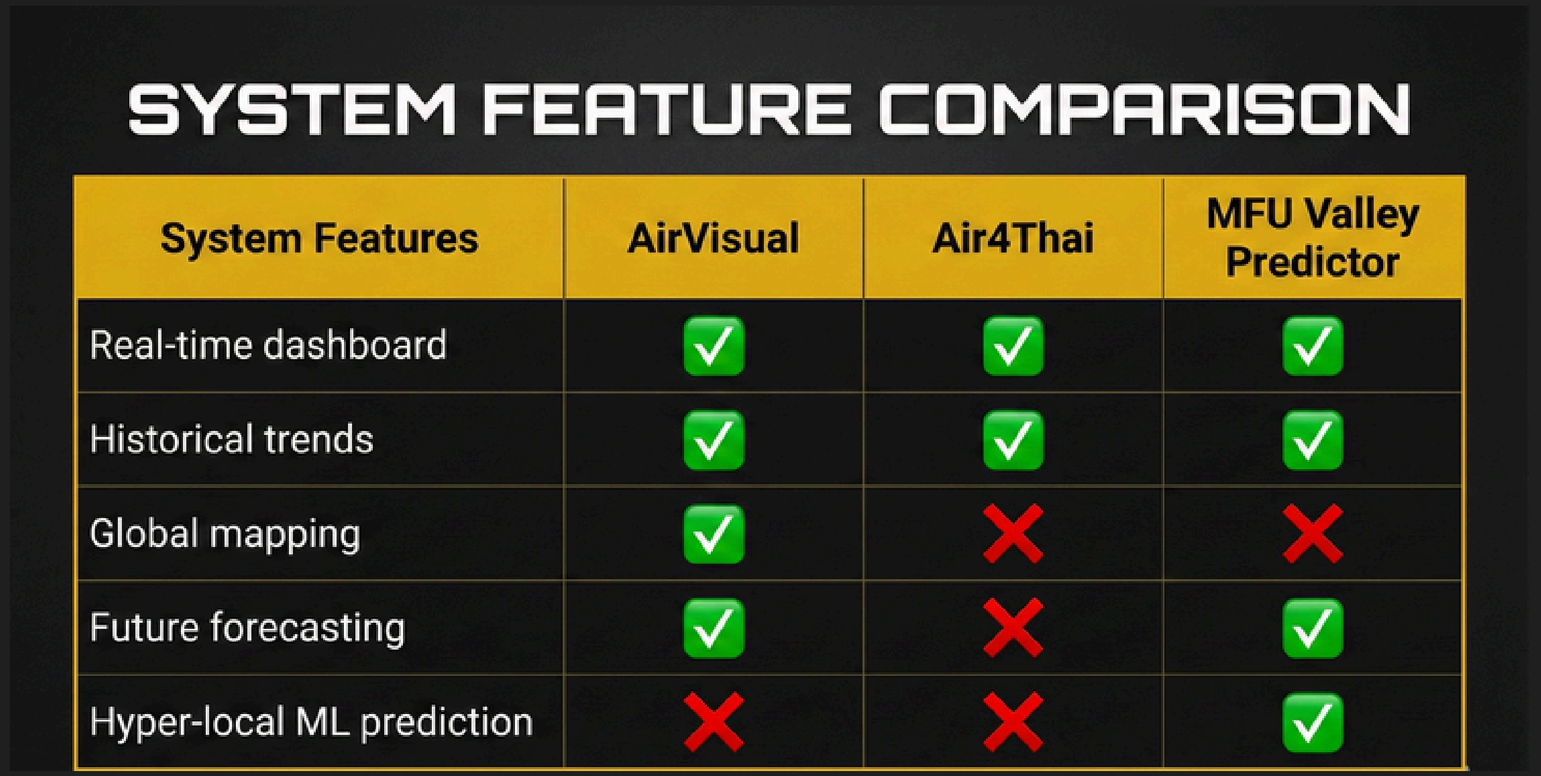
- Depends on Aerosol Optical Depth (AOD).
- Creates significant data "blind spots."
- Fails to track pollution during nighttime.



Local Micro-Climates

- Cannot resolve MFU's specific valley topography.
- Misses sudden local temperature inversions.
- Fails to provide precise local health warnings

System Feature Comparison : The Gap

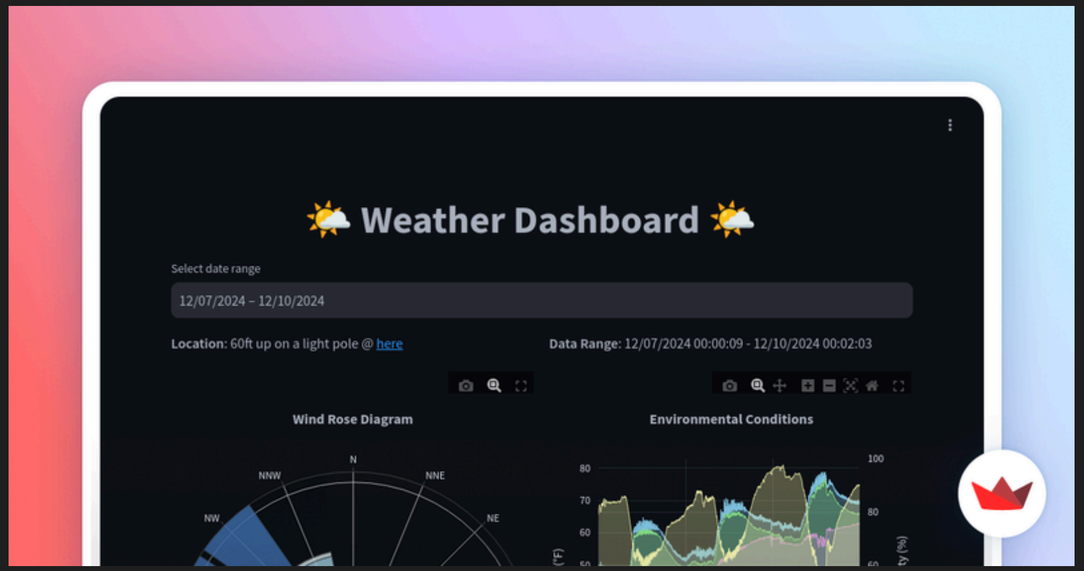


THE GAP: Global tools report the present. MFU requires a local ML model to predict the future of our specific micro-climate changes and bowl effect geography.

Project’s Objective

To build a local, interactive forecasting dashboard specifically calibrated for the MFU valley

Dashboard Mockup



- **Analyze**: Understand how local weather factors interact with MFU's unique topography.
- **Predict**: Forecast localized PM2.5 levels using machine learning.
- **Visualize**: Provide a centralized dashboard where the campus community can easily check predictions and daily activity guidelines.

Technology & Tools

python

- Programming language for building the core backend infrastructure

Pandas & NumPy

- Data manipulation libraries for cleaning and preprocessing Kaggle PM2.5 datasets

scikit-learn

- Used for Data Scaling (***StandardScaler***) and Performance Metrics (R^2, MAE)

XGBoost

- Core predictive engine for high-accuracy PM2.5 forecasting

Streamlit

- Rapid web dashboard development for real-time AQI monitoring

OpenWeatherMap API

- Dynamic retrieval of real-time and 5-day forecasted meteorological data

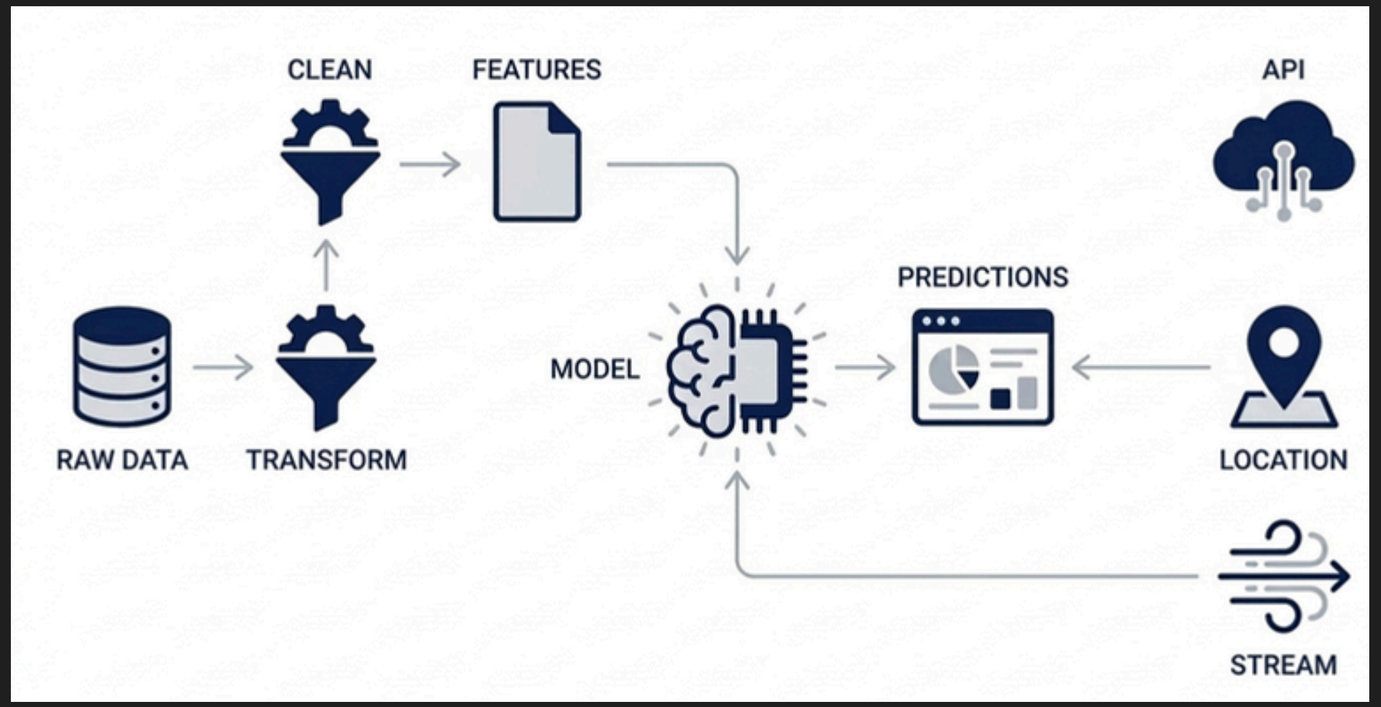
Data Sourcing & Pipeline Strategy

Training Data

- Chiang Mai 7-years-full data
- Cleaned via custom ETL pipeline
- Stored as lightweight CSV

Live Prediction

- OpenWeatherMap & Air4Thai API integration
- Pinpointed to MFU **(Lat: 20.045, Lon: 99.895)**
- Real-time & 5-Day forecast



Model Testing & Features Used

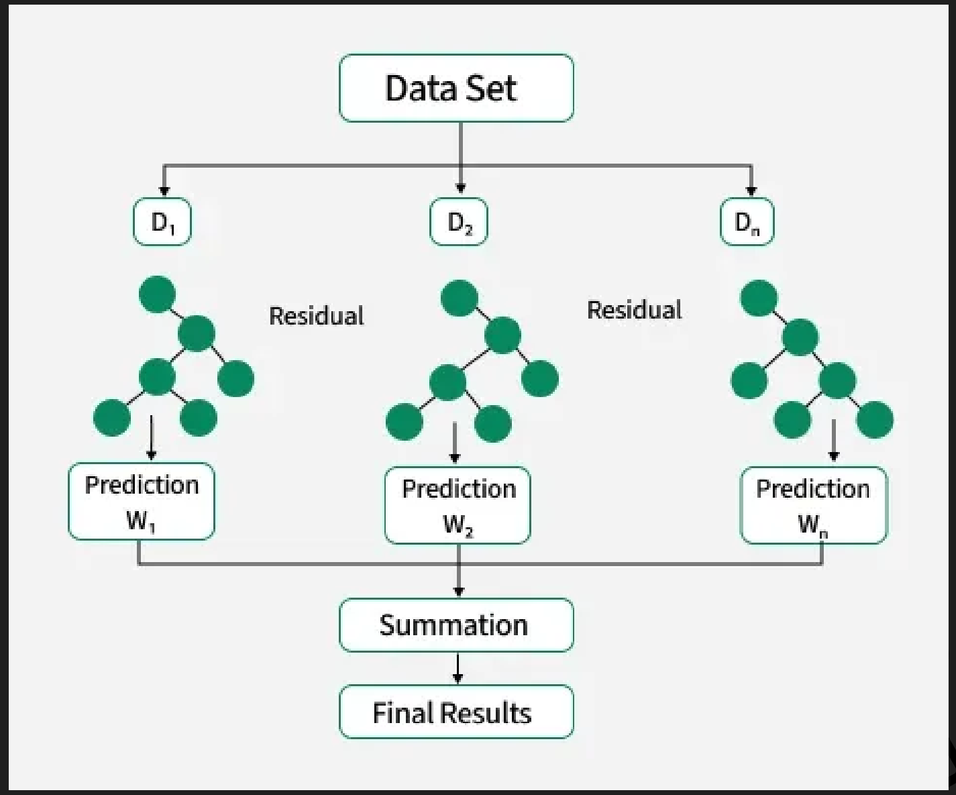
Features Used: 7 Meteorological factors

(*Temperature, Humidity, Wind Speed, Wind Direction, Atmospheric Pressure, Precipitation, Sunshine)*
& One Historical Anchor(pm25_lag1)

Extreme Gradient Boosting (XGBoost) Model

Achieved R^2 Score of 86.52% (MAE: 5.38)

(Outperformed Random Forest baseline and other two models)



Why XGBoost?

- Sequentially corrects previous prediction errors
- Captures complex non-linear relationships in weather data
- Provides higher accuracy compared to traditional models
- Proven superiority over our Random Forest baseline during testing.

Developed Live Web Dashboard Features

User-Centric Features for Proactive Air Quality Monitoring

- **Real-Time MFU Air Quality: Display** live, localized PM2.5 data.
- **5-Day Predictive Forecast**: **Predict** future trends using our XGBoost engine.
- **AQI Health Guidance: Provide** color-coded, actionable health alerts.
- **Live Model Comparison: Contrast** predictions across 4 different models (XGBoost, RF, SVR, MLR) visually.



Project Scope Summary

Senior Project I (Completed Tasks)

- Designing dynamic data pipeline using OpenWeatherMap API
- Structuring 3-Tier Architecture (Presentation, Logic, Data Tiers)
- Training XGBoost model for accurate PM2.5 prediction (Targeting R-squared above 80%)
- Prototyping Streamlit web dashboard for real-time visualization

Senior Project II

(Future Work)





THANK YOU FOR YOUR TIME

“We would like to sincerely thank our advisor and committee members for their support, guidance, and for listening to our project presentation.”

Let us move to Demo! We would love to share our work.